Epitaxial Crystallization Kinetics of Polymer Thin Films on a Solid Substrate

OURSIP SRC Final Research Report

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Abstract

Polymer crystallization of organic semiconductor thin films on solid substrates results in the formation of a wide range of morphologies and microstructures that impact the physical properties of the film, such as its electronic mobility and electrical conductivity. Having a better understanding of the physical mechanisms that affect the crystallization kinetics of polymer films is crucial to improved control of material properties and can lead to the development of new technologies, such as flexible electronics devices. In this summer project, we used a vector-valued diffuse-interface phase field model adapted from [1] to run large-scale computational simulations that studied the effects of thin film diffusivity and mobility anisotropy on observable kinetics parameters, such as growth front velocity, epitaxial needle width, and needle spacing. Our results show that increased mobility anisotropy results in an inverse square root law decrease in growth front velocity, and diffusivity mainly drives crystal growth at low diffusivities. We also found that needle spacing increased in step-function intervals corresponding to the merging of neighboring needles, while needle width largely remained constant throughout the length of the needle. The results of this project provide further insight into the physical behavior of polymer epitaxial crystallization and can be used in the future to derive an exact phenomenological relationship between thin film parameters and crystal properties.

1 Project Context & Motivation

Materials science is an interdisciplinary field that incorporates physics, chemistry, and engineering to study the physical properties of new and existing materials, particularly solids. Because many current problems in science and engineering can be attributed to the limitations of our understanding of the materials available to us, developments in materials science are intricately linked to the advancement of future technology.

Professor Haataja's research group focuses on computational materials science, combining theory and computer simulations to understand the underlying physics of soft condensed matter systems (soft matter). We define soft matter as solid or solid-like systems that are easily deformed by forces on the magnitude of thermal fluctuations. Since the underlying structure of soft matter systems is much larger than atomic scale, we can neglect quantum mechanical effects. Materials considered as soft matter include liquids, polymers, gels, and many biological materials. Improved understanding of the physical properties of these soft matter materials will enable the development of emergent technologies such as soft robotics, bio-medical implants, and stretchable electronics. In these applications, soft materials are desired because they can be readily stretched and deformed while retaining functionality, and are resistant to fracturing.

In this summer's research project, we studied crystallization kinetics of organic polymer thin films on solid substrates. A thin film is defined as a layer of material ranging from nanometers to micrometers in thickness, and is usually synthesized through the process of deposition on a solid surface called the substrate. The particular type of thin film deposition we are interested in is polymer epitaxial growth under heterogeneous confinement. A thorough explanation of polymer epitaxial crystallization is given in section 2.2, and a diagram of the physical structure of the film is shown in Figure 1. We wish to understand how factors such as the diffusivity and mobility anisotropy of the thin film affect observable kinetics parameters of epitaxial crystal needle growth, such as the crystal growth front velocity, needle spacing, and needle width. In this regard, we build upon previous work in [1], which studied the different crystal morphologies that form in organic semiconductor thin films when anisotropy, densification, and free energy were altered in the system.

In the broader context, organic polymer thin films are a particularly promising candidate material for use in flexible electronics technology. Crystallization of these films on solid substrates influences their electronic mobility and therefore can significantly improve their electrical conductivity. Improved understanding of the physical mechanisms that drive polymer crystallization kinetics and lead polymers to crystallize in different morphologies is crucial to better control of material properties, and will enable the behavior of these materials to be reverse engineered for various applications, such as stretchable electronics devices.

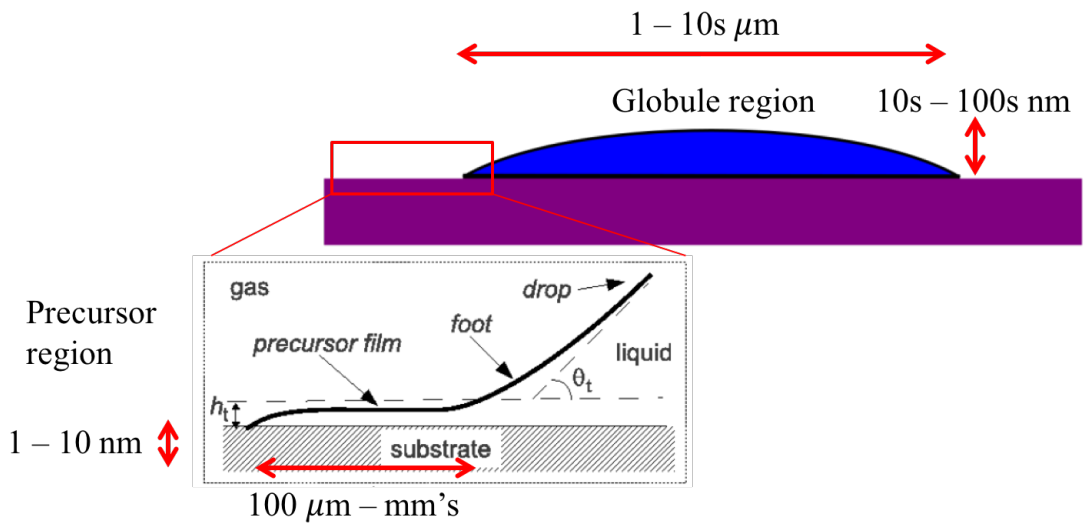

Figure 1: Cross-section diagram of deposition of polymer material onto a solid substrate to form a polymer thin film. Blue area represents the polymer droplet and purple area represents the solid substrate. The inset shows the development of a precursor film layer that is much thinner than the droplet. When viewed from above, the film can be visualized as a "fried egg" model where the droplet is the yolk and the precursor film is the egg white. Diagram by Jason Liu.

2 Background Information

2.1 Theoretical Concepts

The process of crystallization of a solid from an amorphous liquid is an example of a first-order phase transition, which is a phase transition that involves the transfer of latent heat. Specifically, crystallization is a form of solidification. We use phase field theory (PFT) to physically model this phase transition. PFT has its roots in the mean field theory approximation that is used to characterize phase transitions in systems such as the Ising model of ferromagnetism. The theory is dependent on an order parameter, denoted ϕ. The order parameter is usually linked to a microscopic property of the system that characterizes its symmetry; for example, in the Ising model, ϕ denotes the average magnetization of the system, with $+1$ denoting all spins up and -1 denoting all spins down. In the case of polymer crystallization, we use ϕ to denote the crystallinity of our system, with $+1$ denoting crystalline order and 0 denoting amorphous order. Microscopically, in our case, ϕ is related to the Fourier modes that come from modeling the density of a crystalline solid as periodic delta functions; these are also related to the reciprocal lattice vectors encountered in solid-state physics [2].

According to the Landau theory of phase transitions, the Gibbs free energy of the system can be Taylor expanded around the critical transition temperature to yield a simple polynomial expression in ϕ for the free energy. Further, by statistical thermodynamics, all systems

seek to reach equilibrium by minimizing their free energy, so we can differentiate the free energy expression to analytically obtain the order that the system takes below the transition temperature. To incorporate the fact that crystalline and amorphous phases can co-exist spatially, we must also introduce gradients in ϕ. This leads to the introduction of interfacial energy, which is free energy expended to maintain the interface between the two phases in space. The introduction of interfacial energy thus leads to a modification of Landau theory, called the Landau-Ginzberg theory of phase transitions. Under Landau-Ginzberg theory, ϕ becomes a scalar field of the system. Now the free energy becomes a functional with energetic contributions from the interface (which are on the order of $\sim |\phi|^2$) and the bulk (a polynomial in ϕ), which we seek to minimize using the calculus of variations. It is this minimization of the free energy functional that acts as a driving force for how ϕ evolves over time.

Since the crystallizing system is inherently not in equilibrium, ϕ will evolve over time. For crystallization processes, ϕ is not a conserved quantity since the system can go from all amorphous to all crystalline. Therefore, the dynamics are governed by a Langevin-like equation, where $\frac{d\phi}{dt}$ is proportional to the driving force that results from minimization of free energy. Furthermore, stochastic thermal fluctuations that obey the fluctuation-dissipation theorem must be included as well to reflect the fact that the system tends towards equilibrium in the long run. All in all, the dynamics reflect the fact that solidification is driven by the competition between thermodynamics, which controls the undercooling of liquid ahead of the solidification front, and the ability to diffuse latent heat of fusion away from the solid-liquid interface.

Applied specifically to polymer crystallization, this physical competition takes on a slightly different, albeit physically analogous, form. In this case, crystallization is driven by the competition between thermodynamics, which seeks to minimize the free energy necessary to maintain the interface between crystalline and amorphous material, and kinetics, which seeks to diffuse available polymer material as rapidly as possible from the amorphous liquid to the crystal. The physical competition can be illustrated as follows. Nature favors a planar crystallization front because that minimizes the interface surface area. However, diffusion of polymers to make the crystal would be extremely slow because the front must obtain material from further and further away. In particular, the time scale of growth goes like $\sim t^{1/2}$. On the other hand, kinetics favors a needle-like crystallization front because that would allow diffusion of polymers to rapidly increase as it only needs to draw from material near the tip. But nature abhors this morphology because the surface area is huge. So, nature strikes a balance in the competition between thermodynamics and kinetics, favoring a flat and dull needle-like structure for the crystallization front. While this balancing act is known qualitatively, it is precisely in this project that we seek to quantitatively analyze the underlying physical behavior in order to develop a phenomenological theory that will explain the kinetics of epitaxial needle growth of thin-film polymer crystals on solid substrates.

2.2 Experimental & Observational Concepts

As mentioned previously, in this project we study a particular type of thin film deposition known as polymer epitaxial growth under heterogeneous confinement. By epitaxial growth, we refer to deposition of polymer crystals that follows the crystalline structure of the solid substrate. By heterogeneous confinement, we refer to the fact that the thin film material and substrate material are different. An example of the molecular structure of this phenomenon is shown in Figure 2.

When the polymer is first deposited on the substrate, it forms a large globular region where crystalline growth is driven by nucleation. A much thinner precursor film then surrounds the globular region; crystalline growth in this region is driven by diffusion. Emanating from these initial globules are thin epitaxial crystalline needles. Referring to Figure 1, one can imagine the needles as sharp lines of growth emanating from the "egg white", or precursor film, region. While it is known that crystal growth in the droplets is nucleation-limited and crystal growth in the precursor film is diffusion-limited, is currently not known what governs crystal growth of the epitaxial needles.

On the experimental side, we have been collaborating with graduate student Jason Liu of the Arnold Lab (MAE). Jason has been conducting experiments on crystallization of organic polymer thin films using a polyethylene glycol (PEG) polymer on a mica substrate, building upon experimental work in [3]. Specifically, he has been depositing droplets of PEG onto the mica, and then controlling the undercooling of the polymer to induce crystallization. By varying parameters such as the molecular weight, he is able to control the diffusivity of the resulting thin film, allowing us to observe the crystal morphologies that result from such growth. We compared the results of our simulations with observations from his atomic force microscope (AFM) images of polymer thin film crystallization, as shown in Figure 3, to analyze how our simulation results aligned with actual observed data.

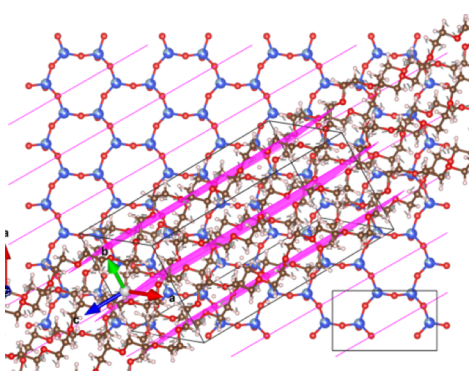

Figure 2: Molecular structure of epitaxial crystallization of polymer material on a solid substrate. Polymer is represented by the brown molecules, with growth direction highlighted by the magenta line. Substrate is represented by the red and blue molecules and is shown to have a quasi-hexagonal structure. Because of the hexagonal structure, favored directions of growth are the ones orthogonal to the sides of the hexagon. Diagram by Jason Liu.

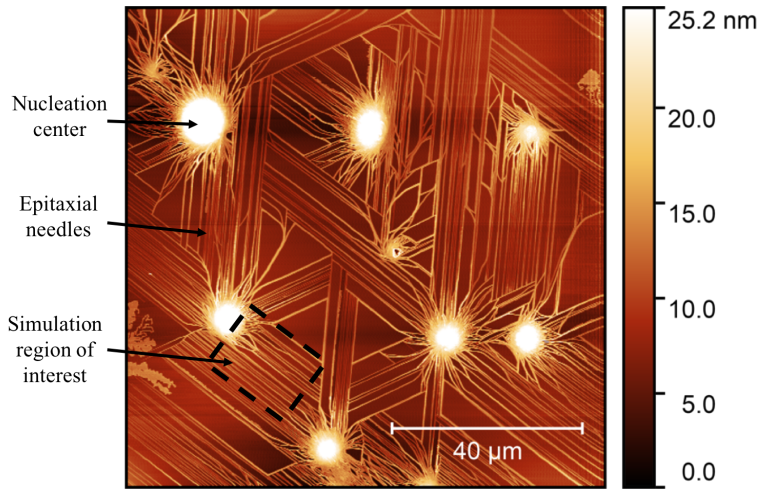

Figure 3: Atomic force microscope (AFM) image of epitaxial crystallization of PEG polymer on a mica-based substrate. The computational simulations we ran in this project focus on the region highlighted by the dashed box. Photo by Jason Liu.

3 Methodology

We used a vector-valued diffuse-interface phase field computational model for polycrystalline solidification, like the one used in [1], and studied how thin film parameters like diffusivity and mobility anisotropy governed the kinetics of polymer crystal growth. We made the order parameter a vector $\vec{\phi}$ so that we could model crystal growth in any three-dimensional direction. Each polymorph was labeled by a 3D vector order parameter $\vec{\phi}_i(\vec{r}, t)$, where i indexes the polymorph. The magnitude of $\vec{\phi}$ was defined such that $|\vec{\phi}| = 1$ (0) represents the crystalline (amorphous) phase. The direction of $\vec{\phi}$ represented the axis of fast crystal growth. We also introduced a scalar order parameter $u(\vec{r}, t)$ which represented the availability of polymers for crystallization at the growth front by diffusion.

The free energy functional F of the system is given by equation 1. It includes the interfacial energy expressed in terms of $\nabla \cdot \vec{\phi}$ and $\nabla \times \vec{\phi}$, the bulk energy term $f_i(\vec{\phi}_i, u) \sim a\phi^4 + b\phi^3 - c\phi^2$, and a term penalizing (increasing free energy) presence of multiple polymorphs at the same location. The parameters a, b, and c are all dependent on θ (angle between x-axis and projection of $\vec{\phi}$ onto xy-plane) and φ (angle between $\vec{\phi}$ and z-axis). The free energy is summed over all N_p polymorphs of the system.

$$F = \sum_{i=1}^{N_p} \int d\vec{r} \left(\frac{\epsilon_i}{2} [(\nabla \cdot \vec{\phi}_i)^2 + |\nabla \times \vec{\phi}_i|^2] + f_i(|\vec{\phi}_i|, u) + a \sum_{j=i+1}^{N_p} |\vec{\phi}_i|^2 |\vec{\phi}_j|^2 \right). \tag{1}$$

The dynamics of $\vec{\phi}$ and u follow usual physical laws. The evolution of $\vec{\phi}$ is given by equation 2. It is dictated by Langevin-like dynamics, with the minimization of the free energy as the driving force. The mobility M, which is the constant of proportionality in equation 2, is

dependent on the mobility anisotropy δ_θ. As stated previously, thermal noise $\vec{\eta}$ is included in equation 2 to obey the fluctuation-dissipation theorem. The evolution of u is given by equation 3. It is dictated by a dimensionless diffusion equation, with an additional factor proportional to $1/\Delta$, where Δ is related to degree of densification during crystallization. Physically, Δ is equal to the ratio of density of amorphous material to density of crystalline material; it essentially controls how diffusion-limited the crystal growth is.

$$\frac{\partial \vec{\phi}_i}{\partial t} = -M_i \frac{\delta F}{\delta \vec{\phi}_i} + \vec{\eta}_i. \tag{2}$$

$$\frac{\partial u}{\partial t} = \nabla^2 u + \sum_{i=1}^{N_p} \frac{1}{\Delta_i} \frac{\partial |\vec{\phi}_i|}{\partial t}. \tag{3}$$

Equations 1, 2, and 3 form the core governing equations of our computational model. Using this model, we ran simulations that modelled a 2-D ultrathin polymer film of dimensionless size $1600 \times 400\,(xy)$. Utilizing dimensionless units throughout our simulations enabled us to investigate the physics of the system so that it could scale for any physical size of the film. Crystallized polymer was represented by $|\vec{\phi}| = 1$, amorphous polymer was represented by $|\vec{\phi}| = 0$, available polymer for crystallization was represented by $u = -1$, and unavailable polymer for crystallization was represented by $u = 0$. The simulation used a Newton-Raphson algorithm to numerically solve the governing partial differential equations at each time step in the simulation. The default values of the parameters we used were: $A = 200$, $D = 12$, and $\epsilon = 10^{-5}$, where A is dimensionless anisotropy (related to δ_θ), D is dimensionless diffusivity (inherent parameter of equation 3), and ϵ is dimensionless interfacial energy (related to ϵ_i in equation 1). For all simulations, we held ϵ constant but varied A from 5 to 1000 and D from 5 to 25.

Each simulation consisted of 200000 time steps, with each time step lasting 3×10^{-7} seconds. Ten snapshots of the simulation were recorded at equal intervals throughout the run to capture the evolution of the crystallization kinetics. For each distinct parameter setting of the simulation, five trials were run. After the simulations ran, we obtained the final values of $|\vec{\phi}|$ and u everywhere in the film. We graphed the results of $|\vec{\phi}|$ in each snapshot as a 2-D contour map, which gave a clear picture of regions of crystallized polymer. Using the contour map of each snapshot, we then wrote several computer programs to calculate the velocity of the crystal growth front, the width of the crystallized needles, and the spacing between the needles. We then used this data to explore trends and determine the analytic relationship between thin film parameters and kinetics observables.

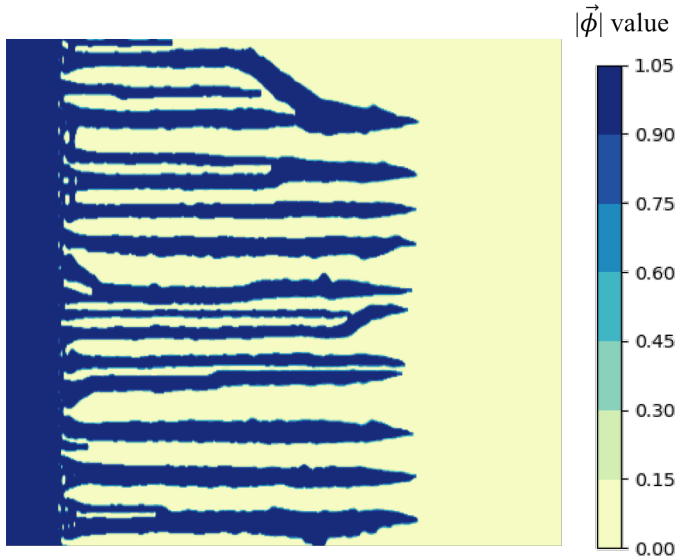

Figure 4: Example of a 2-D contour map created after compiling the results of a simulation run. Blue color represents fully crystallized polymer, yellow color represents amorphous polymer. This contour map comes from the 9th snapshot of the 1st trial of the parameter setting $A = 50$, $D = 12$, where A is dimensionless mobility anisotropy and D is dimensionless diffusivity.

4 Results

4.1 Morphology Characteristics

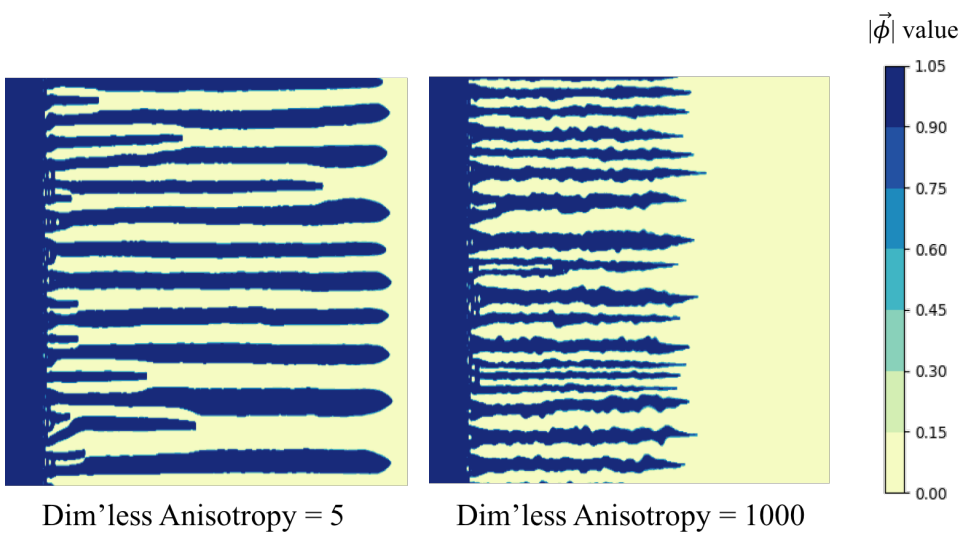

Figure 5: The contour map on the left is from a simulation with $D = 12, A = 5$. The contour map on the right is from a simulation with $D = 12, A = 1000$. Clear differences in the morphology are displayed.

Throughout the range of diffusivity and anisotropy tested, marked changes in morphology were observed, though the contrast is best seen in simulations with extremely high or extremely low anisotropy. As Figure 5 shows, needle morphology for low A is round and smooth, whereas needle morphology for high A is jagged and rough. Furthermore, the low anisotropy needles grew much further in the x-direction than the high anisotropy needles grew. It was also observed from the simulation results that it was quite common for some needles to cut off their neighbors, thus impeding their growth. Hence, although 20 needles can be seen to begin to grow from the nucleation center in the left image of Figure 5, only 10 needles survived to grow to the right end of the film. Additional morphologies that were observed, but not shown here in the report, include needle growths in particular directions such as 60° from the x-axis.

4.2 Crystal Growth Front Kinetics

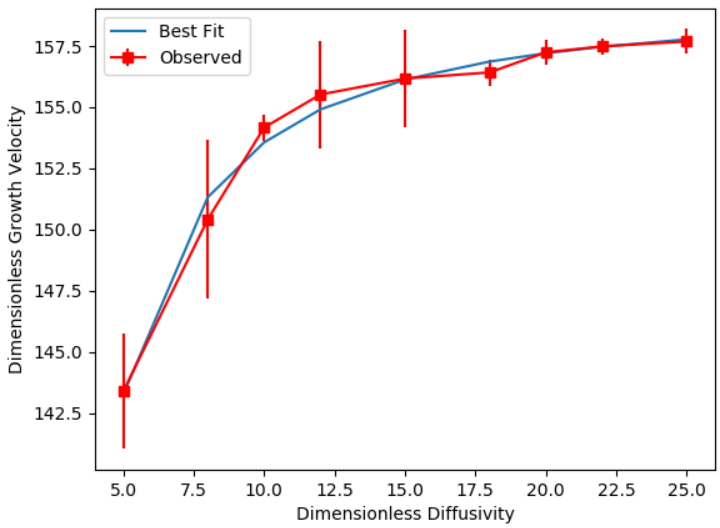

Figure 6: Graph of dimensionless crystal growth front velocity versus diffusivity. Growth front velocity increased for low diffusivities, but plateaued to a constant value around $D = 12$. Using method of least squares, a best fit curve was found for the data.

Crystal growth front kinetics was calculated using an algorithm developed by the author, whereby a peak-fitting algorithm imported from Python was used to determine, to reasonable accuracy, the position of the tips of the needles at each snapshot. The velocity was then calculated by tracking the displacement of the tips over each snapshot. We found that, over a range of dimensionless diffusivities, the dimensionless growth velocity grew until it plateaued around a value of 155.5 ± 2.2 at $D = 12$. By using a curve-fitting algorithm, we calculated the preliminary ansatz for the best-fit curve to be (v is dimensionless growth velocity):

$$v = 159 - \frac{178}{D^{3/2}}. \tag{4}$$

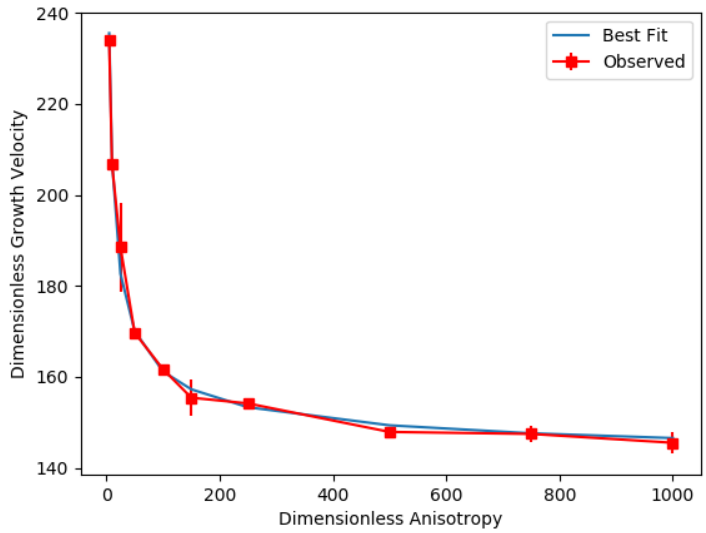

Figure 7: Graph of dimensionless crystal growth front velocity versus anisotropy. Growth front velocity decreased as A increased from 5 to 1000. Using method of least squares, a best fit curve was found for the data.

We also calculated v for trials where we varied A while holding all other parameters constant. The algorithm used was identical to the one discussed above. In the case of anisotropy, we found that growth velocity decreased until stopping around a value of 145, with v changing by only 10 units for $A = 200$ to $A = 1000$. Using the same curve-fitting technique as for the diffusivity trials, we found the preliminary ansatz for the best-fit curve to be:

$$v = 140 + \frac{214}{A^{1/2}}. \tag{5}$$

4.3 Needle Width and Spacing Profiles

Needle width and spacing were calculated using a similar algorithm to the one developed to calculate growth front velocity. After using the peak-finding algorithm in Python to calculate needle tip position, we defined the needle width to be the cross-section length of the needle and the needle spacing to be the distance between peaks. Thus we were able to calculate the spatial variation in the needle width and spacing from the nucleation center to the tip, as well as average over the length of the needle to find the average width and spacing for a particular parameter setting. Our results in Figure 8 show that needle spacing increased as distance from the nucleation center increased, with plateaus indicating merging neighbor needles. Needle width remained largely constant throughout the length of the needle.

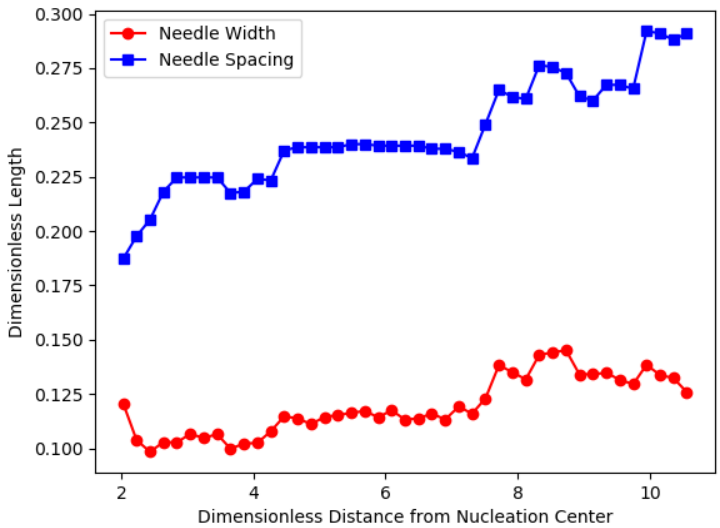

Figure 8: Graph of needle width and spacing versus dimensionless distance from the nucleation center. Data taken from $D = 12$, $A = 50$ simulation run.

5 Conclusions & Future Directions

In this summer project, we used computational phase field model simulations to investigate the crystallization kinetics of organic polymer thin films. Based on our results, we have reached several conclusions which we discuss below.

First, we found significant differences in the morphology of low anisotropy growth versus high anisotropy growth: low anisotropy needles were round and smooth, while high anisotropy needles were smaller and more jagged. The differences suggest that low anisotropy morphology is caused by the lack of preferred growth direction, meaning that material can grow in any direction, whereas high anisotropy morphology is caused by specific growth directions being energetically favored, thus reducing the amount of crystallization that can occur in the x-direction. In addition, the local availability, or lack thereof, of polymer material for crystallization is a contributing factor to the phenomenon of merging neighboring needles that is observed in many of the morphologies. When a needle with high anisotropy reaches an area of depleted polymer availability, the needle usually finds it favorable to turn and merge with a neighboring needle.

Second, we found that diffusivity drives crystal growth at low dimensionless diffusivity, with much reduced effects on crystal growth at higher diffusivities. It is hypothesized that crystal growth at higher D is more influenced by thermal undercooling than diffusivity. According to Figure 6, growth velocity steadily increased as the diffusivity until the cutoff value of $D = 12$. The calculated best-fit ansatz in equation 4 confirms this behavior. Yet, it is important to point out that this ansatz only holds for the current diffusivity parameter range. It cannot be extrapolated to all diffusivities because the behavior of the ansatz in

the limit $D \to \infty$ is incorrect, as such a case should lead to $v \to \infty$. Regarding the effects of anisotropy on growth velocity, we conclude that high anisotropy films rapidly take away material for crystal growth in the horizontal direction and redistribute it to grow in the particular favored directions of growth, thereby reducing the calculated growth velocity. Because low anisotropy films do not have such favored directions, their crystal growth is much less unimpeded in the horizontal direction, leading to higher growth velocity. It was observed that $A = 200$ is the approximate cutoff where increases in anisotropy beyond that value do not significantly affect the growth velocity.

Finally, we found that while needle width largely remained constant throughout the length of the needle, the needle spacing increased in a step function manner, with increases to a higher plateau indicating that neighboring needles had merged. It is interesting to note that although neighboring needles can merge, their width did not change very much after the merger. The cause of the needle merger is likely due to the depletion of available polymer for crystal growth, as mentioned earlier in the conclusion.

In the future, we would like to investigate the effects of additional thin film parameters such as interfacial energy and spatial variation of the densification profile on the crystallization kinetics. We would also like to extend the range of our current parameters in future simulations to A ranging from 5 to 2000 and D ranging from 5 to 100, in order to investigate crystallization kinetics at more extreme parameter values. In addition to these areas, it would also be of interest to calculate the average radius of the needle tip by fitting a parabolic function to each tip, and comparing the simulation results with the theoretically predicted values based on the theory developed in [4].

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Acknowledgements

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